

Schizandra chinensis and *Scutellaria baicalensis* Counter Stress Behaviors in Mice

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The individual and combined antistress effects of the fruit of *Schizandra chinensis* and the radix of *Scutellaria baicalensis* were evaluated using a mouse acute stress model. Stress consisted of immobilization and electric foot shocks over 5 days. Mice were treated with herbal extracts for 7 days before exposing the animals to stress. Before each stressor presentation, the mice were treated with each herbal extract. Reduced locomotor activity and the percentage of time spent in the open arms of an elevated plus-maze under stress were recovered by treatment with the extract containing equal amounts of *S. chinensis* and *S. baicalensis* (CB11) at 200 and 400 mg/kg ($p < 0.05$). The effects of CB11 were greater than the effects of *S. chinensis* or *S. baicalensis* alone. CB11 treatment (100, 200 and 400 mg/kg) significantly reduced serum corticosterone levels ($p < 0.05$). Spleen size and the serum interleukin-2 level decreases induced by stress were prevented by CB11 (200 mg/kg) ($p < 0.05$). Taken together, these results suggest that *S. chinensis* and *S. baicalensis* in equal amounts could be used to treat stress disorders, in part, by preventing corticosterone and IL-2 level changes and ameliorating stress-related behavior parameters. Copyright © 2007 John Wiley & Sons, Ltd.

Keywords: immobilization; electric foot shock; *Schizandra chinensis*; *Scutellaria baicalensis*; interleukin; corticosterone.

INTRODUCTION

Stress reduces physical and mental performance and may induce the progression of existing illness or unmask latent disorders. Stress-derived disorders such as insomnia, anxiety and neurosis have increased in recent years (Gold and Chrousos, 2002). Preclinical and clinical evidence indicates immune functions may be altered by stress (Kook *et al.*, 1995; Stein *et al.*, 1991). It is well known that the immune and central nervous system (CNS) are strongly interrelated. Pro-inflammatory cytokines are increased by stress (Haack *et al.*, 1999), presumably due to the dysregulation of the cytokine milieu (Anisman *et al.*, 1999; Muller and Ackenheil, 1998). Of these cytokines, interleukin-2 (IL-2), which is released from activated T cells, is partly involved in the communication between the immune system and CNS.

On the other hand, corticosteroid hormones operating in concert with catecholamines and other transmitters play important roles in the control of stress reactions. For example, if corticosteroid control is insufficient, stress reactions are enhanced (Heim *et al.*, 2000). It has been reported that antistress drugs also appear to have a corticosteroid sparing effect (Messaoudi *et al.*, 2005). Many herbal medicines such as Ginseng and *Withania somnifera* show significant antistress activity in animal models of experimental stress and have immunomodulatory actions (Kaneko and Nakanish, 2004; Rege *et al.*, 1999).

Schizandra chinensis Turcz. (Schizandraceae) is an important therapeutic herb, and several studies have found that the fruit of *S. chinensis* has hepatoprotective and neuroprotective effects (Mizoguchi *et al.*, 1991; Kim *et al.*, 2004). However, the antistress effects of *S. chinensis* remain debatable. Panossian *et al.* (1999) reported that the increased salivary NO and cortisol in athletes were lowered by *S. chinensis* and *Bryonia alba* extract treatment. However, cold stress-induced gastric ulcers were not ameliorated by the *S. chinensis* extract (Hernandez *et al.*, 1988).

On the other hand, *Scutellaria baicalensis* Georgi. (Labiatae) is an important herb in traditional Chinese medicine. Several investigations have indicated that the radix of *S. baicalensis* has an activity profile that is consonant with putative antioxidant activity. *S. baicalensis* and baicalin (a major flavone constituent of *S. baicalensis*) have many pharmacological properties, e.g. antiinflammatory and antioxidant activities (Gao *et al.*, 1999; Piao

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et al., 2004). Although acute anxiolytic effects have been reported, their effects on stress were not proven (Paladini et al., 1999; Liao et al., 2003). The anxiolytic and immune effects of *S. chinensis* and *S. baicalensis* persuaded us to study their antistress activities.

The present study investigated the antistress effects of extracts of *S. baicalensis* and *S. chinensis* or of their combination on the acute stress model in mice. The antistress effects were evaluated using behavioral parameters and biochemical measures, namely, serum corticosterone and IL-2 concentrations. Restraint stress and electric foot shock were employed to induce a stress burden. The behavioral changes induced by acute stress were assessed using the elevated plus-maze (EPM).

MATERIALS AND METHODS

Materials. Baicalin, corticosterone and dexamethasone were purchased from Sigma Chemical Co. (USA). Schizandrine was purchased from Wako Chemical Co. (Japan). Diazepam was donated by DaeWon Pharmaceutical Co. (Korea). Dried radix of *S. baicalensis* and fruit of *S. chinensis* were purchased at the Kyungdong Oriental drug store (Seoul) and voucher specimens (*S. chinensis*; KHUOPS2004-14, *S. baicalensis*; KHUOPS2004-15) were deposited at the herbarium of the College of Pharmacy at Kyung Hee University. The material was authenticated by Emeritus Professor Chang Soo Yook of the Department of Oriental Pharmaceutical Science, College of Pharmacy, Kyung Hee University. All other materials were of the highest grade commercially available.

Animals. Male ICR mice, weighing 25–30 g, were purchased from the Orient Co., Ltd. (a branch of Charles River Laboratories; Seoul). Animals were housed 4 or 5 per cage, allowed free access to water and food, and maintained under constant temperature ($23 \pm 1^\circ\text{C}$) and humidity ($60 \pm 10\%$) under a 12-h light/dark cycle (light on 07:30–19:30 h). Animal treatment and maintenance were conducted in accordance with the Principles of Laboratory Animal Care (NIH publication No. 85-23, revised 1985) and with the Animal Care and the Use Guidelines of Kyung Hee University, Korea.

Herbal sample preparation. The water extracts of *S. chinensis* and *S. baicalensis* were prepared by boiling dried fruit of *S. chinensis* and/or radix of *S. baicalensis* in 10 volumes of water for 2 h. The aqueous solutions so obtained were filtered, concentrated on a water bath under vacuum, frozen, lyophilized (Eyela, model FDU-2000, Japan) and stored at -20°C until required. In the case of the combined *S. chinensis* and *S. baicalensis* extract, each extract was prepared by extracting mixtures (w/w) of *S. chinensis* and *S. baicalensis* in the following ratios: 1:9 (CB19), 1:2 (CB12), 1:1 (CB11), 2:1 (CB21) and 9:1 (CB91), respectively. The extract yields were as follows: *S. chinensis* extract; 29.1%, *S. baicalensis* extract; 31%, CB19; 38.9%, CB12; 36.7%, CB11; 36.4%, CB21; 36.4%, CB91; 41.3%. Schizandrine in *S. chinensis* and baicalin in *S. baicalensis* were used to ensure preparation consistencies. Baicalin and schizandrine were present in CB11 at 4% and 0.03%, respectively.

Induction of stress and drug administration. The stress consisted of 30 min immobilization and electric foot shocks over 5 days. The procedure involved restraining the animal in a 50 mL Corning tube, with the animal's nose at the normally enclosed end of a tube; ventilation was provided by a hole at the end. Following immobilization stress for 30 min, the mouse was exposed to electric foot shocks with 1 mA (1 s duration with 20 s inter-shock interval) for 5 min daily. Saline-treated control group (hereafter the unstressed control group) were not exposed to stress and treated with saline orally. Saline treated stressed controls (hereafter the stressed control group) were treated with saline and exposed to restraint and electric foot shock stress. Stressed animals were treated daily with herbal extracts using various dosage regimens for 7 days before exposure to stressors. During the stress burdening period, mice were exposed to stress 1 h after receiving herbal extracts. Herbal extracts were freshly dissolved in saline and administered per os (p.o.).

Spontaneous behavior in the open field test. Testing was conducted in clear black Plexiglas boxes ($40 \times 40 \times 40$ cm) equipped with a video-base Ethovision System (Noldus, Wageningen, The Netherlands). To evaluate horizontal locomotor activity, 5 min after terminal stress mice were placed in the center of the apparatus and video-recorded for 5 min. Horizontal locomotor activity was defined as total ambulatory distance.

Elevated plus-maze test. Mice were placed in the EPM 10 min after the final stress session and tested individually once for 5 min. The EPM consisted of two open arms (30×6 cm) and two enclosed arms (30×6 cm) with 30 cm high walls, extending from a central platform (6×6 cm). The arms were connected to a central square, 6×6 cm, giving the apparatus a plus sign appearance. The maze was raised to 50 cm above floor level in a dimly lit room (20 Lux) and a video camera was suspended above the maze to record movements for analysis. The maze floor and walls were constructed from dark opaque polyvinyl plastic. Mice were placed individually at the center of the platform facing an enclosed arm. The maze was cleaned following each trial to remove any residues or odors. The following measurements were taken using the video-based Ethovision System and analysed; the number of entries into open and closed arms, the time spent in each arm, and the total distance moved in the EPM. Diazepam (5 mg/kg, p.o.) was used as a positive control.

Measurements of serum corticosterone and IL-2 levels. After monitoring behavioral activity, as described above, blood samples were taken by heart puncture, timed for between 10:00 a.m. and 2:00 p.m. to avoid diurnal variations, and the spleens were removed and weighed. Serum corticosterone levels were determined using a modification of a previously described method (Harikai et al., 2003) using an HPLC system composed of an SI-2 3001 pump, an SI-2 3002 UV-visible detector, an SI-2 3004 column oven, separation column (Shiseido, Tokyo, Japan), and column Capacell Pak C18 MG 120 ($5\mu\text{m}$, 3×250 mm). Dexamethasone was used as the internal standard. Corticosterone levels were determined by measuring absorbance in 240 nm using dsCHROM-computing program (Shiseido, Tokyo, Japan).

Serum IL-2 levels were measured using an ELISA kit (R&D systems, MN, USA).

Statistics. Values are expressed as mean $\pm$ SEM. Data were analysed by one-way analysis of variance (ANOVA) followed by the Student-Newman-Keuls test for multiple comparisons. Statistical significance was set at $p < 0.05$.

RESULTS

Effect of *S. chinensis* and *S. baicalensis* extracts on spontaneous locomotor activity

Locomotor activities were significantly affected by stress. Moreover, the stressed control group showed a significant reduction in the total distance traveled versus the unstressed control group (1801.1 ± 148.7 cm vs 2344.7 ± 111.6 cm; $p < 0.01$). *S. chinensis* extract (200 mg/kg) markedly increased this total ambulatory distance (2257.1 ± 145.1 cm). *S. chinensis* and *S. baicalensis* combined extracts in all ratios also showed significant increases in total ambulatory distances versus the stressed controls, and in particular, CB11 (200 mg/kg) (*S. chinensis* and *S. baicalensis* at 1:1 (w/w)) showed a marked increase in the total distance traveled compared with the stressed controls (2378.1 ± 62.8 cm vs 2344.7 ± 111.6 cm; $p < 0.01$). However, *S. baicalensis* extract at 200 mg/kg produced no increase in locomotor activity versus the stressed controls (1975.9 ± 41.6 cm vs 1801.1 ± 148.7 cm).

Effect of *S. chinensis* and *S. baicalensis* on EPM performance

Animals exposed or not exposed to stress were significantly different in terms of the times spent in the open or closed arms of the EPM over 5 min. As shown in Fig. 1, stressed controls showed a significant decrease in the percentage of time spent in open arms compared to unstressed controls ($p < 0.001$). However, *S. baicalensis* extract treated mice spent significantly more time in the open arms than the stressed controls ($p < 0.05$). Moreover, mice treated with CB11 and CB91 also showed significant increases in the percentage of time spent in open arms compared with stressed controls. The most effective combination found was CB11. Thereafter, CB11 was focused on and its usefulness examined on the effects of stress.

Effect of CB11 on locomotor activity and EPM performance

As observed above, stressed controls showed a significant decrease in total ambulatory distance versus unstressed controls (2328.4 ± 175.7 cm versus 1729.7 ± 150.6 cm, $p < 0.05$). However, this reduced locomotor activity was prevented by CB11 at 100, 200 and 400 mg/kg (2223.4 ± 96.9 cm, 2319.6 ± 59.6 cm and 2297.2 ± 15.6 cm, respectively). In animals not subjected to stress, it was observed that CB11 produced no significant change in total ambulatory distances at any doses (data not shown). Diazepam-treated mice showed no significant differences in the locomotor activity versus the

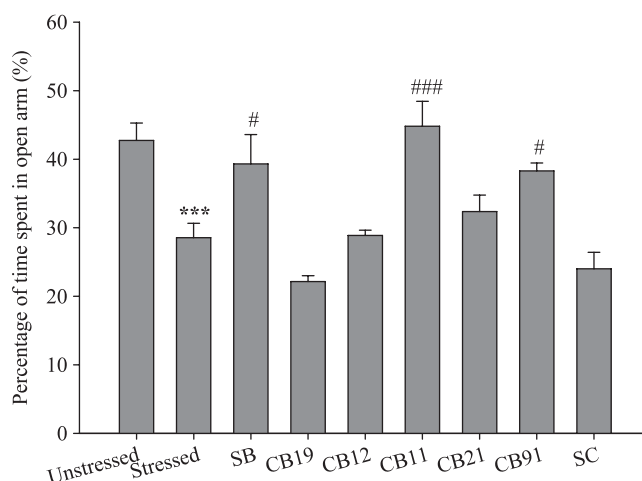


Figure 1. Effects of mixed extracts of the fruit of *Schizandra chinensis* (SC) and the radix of *Scutellaria baicalensis* (SB). Extracts were mixed in the following ratios of *S. chinensis* and *S. baicalensis*; 1:9 (CB19), 1:2 (CB12), 1:1 (CB11), 2:1 (CB21) and 9:1 (CB91). These were treated at 200 mg/kg/day (p.o.). The final treatment was administered 1 h before the elevated plus-maze testing ($n = 8$). Data are expressed as mean $\pm$ SEM of the percentage of time spent in the open arms of an elevated plus-maze for a 5 min test period. Values of p for group comparisons were obtained by one-way ANOVA followed by the Student-Newman-Keuls test (***) $p < 0.001$ versus the unstressed controls, # $p < 0.05$; ### $p < 0.001$ versus stressed controls).

stressed controls (1930.3 ± 155.1 cm versus 1729.7 ± 150.6 cm).

In the EPM test, the stressed controls spent more time in the closed arm than the unstressed controls ($p < 0.01$; Fig. 2). CB11 (200 mg/kg) treated mice showed a significant increase in the percentage of time spent in the open arms than stressed controls ($p < 0.05$; Fig. 2). However, unlike the locomotor activity results, diazepam (5 mg/kg)-treated stressed animals showed a significant increase in the percentage of time spent in open arms versus the stressed controls ($p < 0.05$).

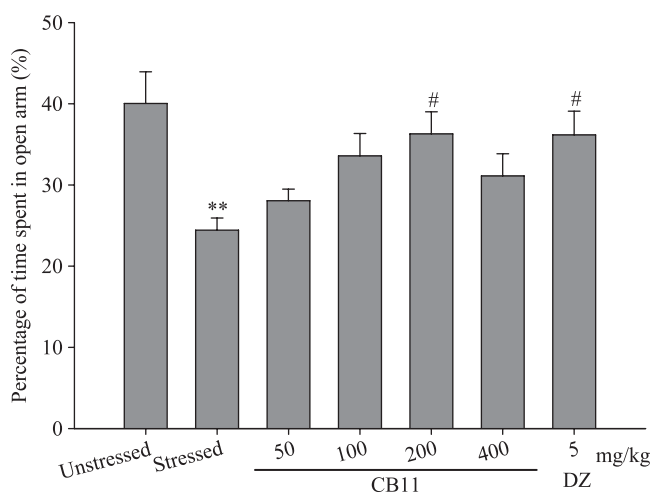


Figure 2. Effects of CB11 (an extract of 1:1 ratio of the fruit of *Schizandra chinensis* and the radix of *Scutellaria baicalensis*) on the elevated plus-maze test ($n = 8$). The extract was treated at 200 mg/kg/day (p.o.). The last treatment was terminated 1 h before the testing. Data are expressed as mean $\pm$ SEM of the percentage of time spent in the open arms of an elevated plus-maze over a 5 min test period. Values of p for group comparisons were obtained by one-way ANOVA followed by the Student-Newman-Keuls test (** $p < 0.001$ versus the unstressed controls, # $p < 0.05$ versus stressed controls). DZ, diazepam.

Table 1. Effects of a combination of the fruit of *Schizandra chinensis* and the radix of *Scutellaria baicalensis* extract (CB11; 1:1 ratio (w/w)) on stress-induced changes in serum corticosterone and interleukin-2 levels, and on spleen weight

	Weight of spleen (mg)	Corticosterone ($\mu\text{g/mL}$)	Interleukin-2 (pg/mL)
Unstressed	127.4 $\pm$ 8.0	59.5 $\pm$ 18.0	44.6 $\pm$ 1.4
Stressed control	102.2 $\pm$ 4.9 ^a	371.8 $\pm$ 63.6 ^b	36.7 $\pm$ 0.6 ^b
CB11			
50 mg/kg	114.0 $\pm$ 4.6	309.5 $\pm$ 34.1	n.d.
100 mg/kg	120.4 $\pm$ 6.4	258.3 $\pm$ 14.9 ^c	n.d.
200 mg/kg	129.7 $\pm$ 5.0 ^c	213.0 $\pm$ 22.4 ^d	42.1 $\pm$ 0.8 ^d
400 mg/kg	109.0 $\pm$ 2.9	244.3 $\pm$ 18.5 ^c	n.d.
Diazepam	101.9 $\pm$ 4.9	195.2 $\pm$ 16.9 ^d	n.d.

The unstressed group was not exposed to any stress, whereas the stressed control group was exposed to stress but not treated with CB11. Data represent mean $\pm$ SEM of 8 mice. The preparation of animals and specimens are described in Materials and Methods. Values of p for group comparisons were obtained by one-way ANOVA followed by the Student-Newman-Keuls test (^a $p < 0.05$; ^b $p < 0.001$ versus the unstressed controls, ^c $p < 0.05$; ^d $p < 0.01$ versus stressed controls). n.d., not determined.

Effect of CB11 on biochemical parameters and spleen weights

To investigate whether changes in biochemical parameters such as serum corticosterone and IL-2 levels after stress are normalized or prevented by CB11 treatment, corticosterone and IL-2 concentrations were measured in mice treated with CB11 (Table 1). In the unstressed controls, the mean serum corticosterone level was $59.5 \pm 18.0 \mu\text{g/mL}$, and this increased significantly in stressed controls ($p < 0.001$). Increased corticosterone levels after stress were significantly reduced by CB11 (at 100, 200 and 400 mg/kg) versus the stressed controls. Diazepam was found to be similar to CB11 in this respect. However, neither CB11 nor diazepam treatment maintained corticosterone at the unstressed control level. Furthermore, stress affected serum IL-2 levels and spleen shrinkage. As shown in the Table 1, mean spleen size and IL-2 levels were reduced by stress. Spleen sizes were gradually and significantly recovered by CB11 (200 mg/kg) to the normal level. IL-2 levels were significantly higher in animals treated with CB11 (200 mg/kg) than in stressed controls ($p < 0.01$). However, mean spleen size in the diazepam-treated positive controls was similar to that of the stressed controls.

DISCUSSION

The present decade has been referred to as the age of stress (Kaneko and Nakanishi, 2004). The manifestation of stress for any reason, causes deleterious physiological changes, alters normal functioning, decreases labor efficiency, increases fatigue, reduces locomotor activity and may induce mental depression. Simple physical fatigue is easily remedied by rest, but mental stress or a combination of mental and physical fatigue is more difficult to cure. Various attempts have been made to cope with the adverse effects of stress. However, current pharmacological treatments appear to have limited utility. Some herbal materials feature in such treatment because they enable the organism to counteract and adapt to various stressors that can adversely affect the physiological system (Bhattacharya and Muruganandam, 2003).

A variety of stress situations have been employed to investigate the consequences of stress and to evaluate the efficacies of antistress agents, but the lack of consistency of these stress protocols and their biological consequences are a concern. Immobilization is one of the most frequently employed experimental stress models, and produce robust increases in serum corticosterone levels and long-lasting behavioral changes, such as decreased exploratory activity (Albonetti and Farabollini, 1994). Moreover, electric foot shock also increases corticosterone levels and reduces hippocampal brain-derived neurotrophic factor (Scaccianoce *et al.*, 2003). In the present study, it was observed that locomotor activity was reduced and anxiety-like behavior increased in stressed mice. In addition to these behavioral parameters, corticosterone levels were increased and spleen weights decreased. The present results are consistent with previously published data (Djordjevic *et al.*, 2003; Takeuchi *et al.*, 2003).

S. chinensis and *S. baicalensis* are well known herbal materials in traditional Chinese medicine and contain many antioxidant compounds. It has been reported that compounds with antioxidative properties also have antistress effects (Iwai *et al.*, 2004; Matsukawa *et al.*, 1997; Butterweck *et al.*, 2004). Thus, *S. chinensis* and *S. baicalensis* extract might be expected to exhibit antistress activity. *S. chinensis* extract contains lignans which have antioxidant and hepatoprotective properties (Chiu *et al.*, 2002; Mizoguchi *et al.*, 1991). Moreover, *S. chinensis* is an adaptogenic material (Panossian *et al.*, 1999). Despite previous published data, antistress effects, including improvement in behavioral or biochemical parameters, were not observed in *S. chinensis* extract-treated animals. The major constituents of *S. baicalensis* are flavone derivatives, such as, baicalin and wogonin, which have antioxidant and antiinflammatory activities. Wogonin also exhibited anxiolytic activity via positive allosteric modulation of the γ -aminobutyric acid (GABA) receptor complex by interacting with the benzodiazepine site (Hui *et al.*, 2002). Modifications of side chains on flavones have been found to be selective for a number of receptor systems, including the opiate receptor and the GABA receptor (Hui *et al.*, 2002; Thirugnana-sambantham *et al.*, 1993). Therefore, it could be that to some extent the antistress effect of *S. baicalensis* extract is due to the actions of flavones, and that it is

mediated by their antioxidative activities and by interactions with GABA receptor. However, *S. baicalensis* extract ameliorated anxiety-like behavior but failed to prevent reduced exploratory activity after stress, which concurs with previously published data (Ryu *et al.*, 2004; Liao *et al.*, 2003). Thus the antistress effects of *S. chinensis* or *S. baicalensis* extract are not fully dependent on their antioxidative properties.

In the present study, synergistic effects were observed when *S. chinensis* and *S. baicalensis* were combined. In particular, behavioral and biochemical parameters associated with stress were most ameliorated by CB11 extract, which contained *S. chinensis* and *S. baicalensis* in equal amounts. Stress-induced anxiety behavior, which in the present was characterized by a reduced time spent in open arms, was reduced by CB11 treatment. Furthermore, increases in serum corticosterone after stress were also reversed by CB11 treatment in a dose-dependent manner. Restraint stress has been reported to increase serum corticosterone levels (Wang *et al.*, 2002). Increases in serum corticosterone are also induced by electric-shock stress (Sasaki *et al.*, 1995). Thus, elevated corticosterone levels are a common feature of acute stress. Our results suggest that CB11 may effectively block stress-induced corticosterone release.

Spleen size and IL-2 levels were recovered by CB11 treatment. Changes in corticosterone levels, IL-2 levels and spleen sizes induced by stress conditions have been suggested to be linked to changes of immune function (Kubera *et al.*, 1998). Moreover, the serum levels of IL-2 and spleen weights were reported to be reduced by various stressors, including swimming stress, restraint stress, chronic mild stress and electric foot shock (Kubera

et al., 1998; Li *et al.*, 2003; Brenner and Moynihan, 1997). The behavioral and biochemical results of the present study are compatible with previously published data. Immune stimulation induces the proliferation of immune cells in the spleen and IL-2 production. Therefore, we speculate two possible mechanisms underlying spleen shrinkage and reduced serum IL-2 concentrations, a hormone-related mechanism involving corticosterone elevation due to stress, or the direct inhibition of spleen immune function via neuronal innervations (Hori *et al.*, 1995). Although the nature of the precise mechanism whereby CB11 reverses stress-induced alterations in serum corticosterone and IL-2 levels and spleen size was not elucidated, we hypothesize it is via CB11 modulation of the immune and nervous systems. Further studies of these issues are required.

In conclusion, although *S. chinensis* extract or *S. baicalensis* extract attenuated behavioral and biochemical changes induced by stress in mice, equal amounts of *S. chinensis* and *S. baicalensis* in combination, acted synergistically to ameliorate the changes induced by restraint and electric foot shock stress. Although these findings do not provide clinically useful outcome data in humans, they do suggest that CB11 has the potential to be used as a functional food or as a drug to alleviate stress responses.

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